SIMPLIFYING EXPRESSIONS

BASIC ALGEBRA

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GRADING SYSTEM

FINAL GRADE

30%

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SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Algebraic Expressions

Properties of Real Numbers

Exponent Rules

Radical Rules

ALGEBRAIC EXPRESSIONS

An expression made up of **letters** and/or **numbers** with sign(s)

$$\mathbf{2x + 3y - 1 = 15}$$

equation

An expression made up of **letters** and/or **numbers** with sign(s)

$$\mathbf{2x + 3y - 1}$$

VARIABLE

Represents a number with **varying** values

$$2\boldsymbol{x} + 3\boldsymbol{y} - 1$$

CONSTANT

Represents a number with **fixed** value

$$2x + 3y - 1$$

NUMERICAL COEFFICIENT

The constant number that **multiplies** a variable

$$\mathbf{2}x + \mathbf{3}y - \mathbf{1}$$

A combination of variable(s) and constant(s) which is separated by **addition** or **subtraction**

$$\boxed{2x} + \boxed{3y} - \boxed{1}$$

PROPERTIES OF REAL NUMBERS

COMMUTATIVE LAW

Addition

$$a + b = b + a$$

$$2 + 3 = 3 + 2$$

$$5 = 5$$

Multiplication

$$ab = ba$$

$$(2)(3) = (3)(2)$$

$$6 = 6$$

ASSOCIATIVE LAW

Addition

$$a + (b + c) = (a + b) + c$$

$$1 + (2 + 3) = (1 + 2) + 3$$

$$1 + 5 = 3 + 3$$

$$6 = 6$$

Multiplication

$$a(bc) = (ab)c$$

$$(2)(3 \cdot 4) = (2 \cdot 3)(4)$$

$$(2)(12) = (6)(4)$$

$$24 = 24$$

DISTRIBUTIVE LAW OF MULTIPLICATION

$$a(b+c) = ab + ac$$

$$2(3+2) = (2 \cdot 3) + (2 \cdot 2)$$

$$2(5) = (6) + (4)$$

$$10 = 10$$

EXPONENT RULES

PRODUCT RULE

$$x^n \cdot x^m = x^{n+m}$$

$$2^2 \cdot 2^3 = 2^{2+3}$$

$$(2 \cdot 2) \cdot (2 \cdot 2 \cdot 2) = 2^5$$

$$2^5 = 2^5$$

In multiplying two expressions with the same base, **add** the exponents

QUOTIENT RULE

$$\frac{x^n}{x^m} = x^{n-m}$$

In dividing two expressions with the same base, **subtract** the exponents

$$\frac{2^4}{2^2} = 2^{4-2}$$

$$\frac{2 \cdot 2 \cdot 2 \cdot 2}{2 \cdot 2} = 2^2$$

$$2 \cdot 2 = 2^2$$

$$2^2 = 2^2$$

POWER RULE

$$(x^n)^m = x^{n \cdot m}$$

In raising a power to a power, **multiply** the exponents

$$(2^2)^3 = 2^{2 \cdot 3}$$

$$(2 \cdot 2)^3 = 2^6$$

$$(2 \cdot 2)(2 \cdot 2)(2 \cdot 2) = 2^6$$

$$2^6 = 2^6$$

POWER OF ZERO

$$\frac{2^3}{2^3} = 1$$

$$x^0 = 1$$

$$2^{3-3} = 1$$

$$2^0 = 1$$

Any number raised to zero is 1,

except $\mathbf{0^0}$ which is **undefined**

NEGATIVE EXPONENTS

$$x^{-n} = \frac{1}{x^n}$$

Any number raised to a negative power is equal to the **reciprocal** of that number raised to the positive power

$$2^{-2} = \frac{1}{2^2}$$

$$\frac{2^0}{2^2} = \frac{1}{2^2}$$

$$\frac{1}{2^2} = \frac{1}{2^2}$$

FRACTIONAL EXPONENTS

$$x^{\frac{1}{n}} = \sqrt[n]{x}$$

Any number raised to the power of $\frac{1}{n}$ is equal to the ***n-th* root** of that number

$$4^{\frac{1}{2}} = \sqrt[2]{4}$$

$$(2^2)^{\frac{1}{2}} = 2$$

$$2^{\frac{2}{2}} = 2$$

$$2 = 2$$

Distribute an **EXPONENT** over a **PRODUCT**

$$(x \cdot y)^n = x^n \cdot y^n$$

The power of a product is equal to the product of the powers

$$(2 \cdot 3)^2 = 2^2 \cdot 3^2$$

$$(6)^2 = 4 \cdot 9$$

$$36 = 36$$

$$(x + y)^n \neq x^n + y^n$$

$$(x - y)^n \neq x^n - y^n$$

Distribute an **EXPONENT** over a **PRODUCT**

$$(x \cdot y)^n = x^n \cdot y^n$$

The power of a product is equal to the product of the powers

$$(2 \cdot 3)^2 = 2^2 \cdot 3^2$$

$$(6)^2 = 4 \cdot 9$$

$$36 = 36$$

$$(2 + 3)^2 \neq 2^2 + 3^2$$

$$(5)^2 \neq 4 + 9$$

$$25 \neq 13$$

Distribute an **EXPONENT** over a **QUOTIENT**

$$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$$

The power of a quotient is equal to the quotient of the powers

$$\left(\frac{1}{2}\right)^2 = \frac{1^2}{2^2}$$

$$(0.5)^2 = \frac{1}{4}$$

$$0.25 = 0.25$$

Simplify the following expression:

$$\frac{3x^{-2}}{x^4}$$

$$\frac{3\,\frac{1}{x^2}}{x^4}$$

Simplify the following expression:

$$\frac{3x^{-2}}{x^4}$$

$$\frac{3\frac{1}{x^2}}{x^4}$$

Simplify the following expression:

$$\frac{3x^{-2}}{x^4}$$

$$3\frac{\frac{1}{x^2}}{x^4}$$

$$3\frac{1}{x^2}\cdot\frac{1}{x^4}$$

$$\frac{3}{x^2x^4}$$

$$\frac{3}{x^{2+4}}$$

$$\boxed{\frac{3}{x^6}}$$

ans

Simplify the following expression:

$$\frac{4y^3}{y^{-5}}$$

Simplify the following expression:

$$\frac{4y^3}{y^{-5}}$$

$$\frac{4y^3}{\frac{1}{y^5}}$$

Simplify the following expression:

$$\frac{4y^3}{y^{-5}}$$

$$\frac{4y^3}{\frac{1}{y^5}}$$

$$4y^3 \cdot \frac{y^5}{1}$$

$$4y^3 \cdot y^5$$

$$4y^{3+5}$$

$$\boxed{4y^8}$$

ans

Simplify the following expression:

$$\frac{y^3 z^5}{7z^{-2}y^7}$$

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$$\frac{y^3 z^5}{7z^{-2}y^7}$$

Simplify the following expression:

$$\frac{y^3 z^5 z^2}{7 \quad y^7}$$

Simplify the following expression:

$$\frac{y^3 z^5 z^2}{7 \quad y^7}$$

Simplify the following expression:

$$\frac{z^5 z^2}{7y^{-3}y^7}$$

$$\frac{z^{5+2}}{7y^{-3+7}}$$

$$\boxed{\frac{z^7}{7y^4}}$$

ans

RADICAL RULES

RADICAL OF A PRODUCT

$$\sqrt[n]{a \cdot b} = \sqrt[n]{a} \cdot \sqrt[n]{b}$$

The radical of the product is the product of the radicals

$$\sqrt{4 \cdot 9} \quad = \sqrt{4} \cdot \sqrt{9}$$

$$\sqrt{36} \quad = 2 \cdot 3$$

$$6 = 6$$

$$\sqrt[n]{a + b} \neq \sqrt[n]{a} + \sqrt[n]{b}$$

$$\sqrt[n]{a - b} \neq \sqrt[n]{a} - \sqrt[n]{b}$$

RADICAL OF A PRODUCT

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The radical of the product is the product of the radicals

$$\sqrt{4 \cdot 9} \quad = \sqrt{4} \cdot \sqrt{9}$$

$$\sqrt{36} \quad = 2 \cdot 3$$

$$6 = 6$$

$$\sqrt{4 + 9} \neq \sqrt{4} + \sqrt{9}$$

$$\sqrt{13} \neq 2 + 3$$

$$\sqrt{13} \neq 5$$

RADICAL OF A QUOTIENT

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

The radical of the quotient is the quotient of the radicals

$$\sqrt[3]{\frac{64}{8}} = \frac{\sqrt[3]{64}}{\sqrt[3]{8}}$$

$$\sqrt[3]{8} = \frac{4}{2}$$

$$2 = 2$$

FRACTIONAL EXPONENTS

$$a^{m/n} = \left(\sqrt[n]{a}\right)^m$$

Any number raised to a fractional exponent is equal to the **root** indicated by the **denominator** of the base raised to the power of the numerator

$$27^{\frac{2}{3}} = \left(\sqrt[3]{27}\right)^2$$

$$\left(27^{\frac{1}{3}}\right)^2 = (3)^2$$

$$\left(\sqrt[3]{27}\right)^2 = 9$$

$$(3)^2 = 9$$

$$9 = 9$$

Compute $(25)^{-3/2}$

$$\frac{1}{(25)^{3/2}}$$

Compute $(25)^{-3/2}$

$$\frac{1}{(25)^{3/2}}$$

$$\frac{1}{\left(\sqrt{25}\right)^{3}}$$

$$\frac{1}{(5)^{3}}$$

$$\boxed{\frac{1}{125}}$$

ans

Simplify the expression

$$\left(\frac{25x^4y^{-5}}{x^{-6}y^3}\right)^{3/2}$$

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Simplify the expression

$$\left(\frac{25x^4}{x^{-6}y^3y^5}\right)^{3/2}$$

Simplify the expression

$$\left(\frac{25x^4}{x^{-6}y^3y^5}\right)^{3/2}$$

Simplify the expression

$$\left(\frac{25x^4x^6}{y^3y^5}\right)^{3/2}$$

$$\left(\frac{25x^{4+6}}{y^{3+5}}\right)^{3/2}$$

$$\left(\frac{25x^{10}}{y^8}\right)^{3/2}$$

Simplify the expression

$$\left(\frac{25x^4x^6}{y^3y^5}\right)^{3/2}$$

$$\left(\frac{25x^{4+6}}{y^{3+5}}\right)^{3/2}$$

$$\left(\frac{25x^{10}}{y^8}\right)^{3/2}$$

$$\frac{25^{\frac{3}{2}}x^{10\cdot\frac{3}{2}}}{y^{8\cdot\frac{3}{2}}}$$

$$\frac{25^{\frac{3}{2}}x^{\frac{30}{2}}}{y^{\frac{24}{2}}}$$

$$\frac{25^{\frac{3}{2}}x^{15}}{y^{12}}$$

Simplify the expression

$$\left(\frac{25x^4x^6}{y^3y^5}\right)^{3/2}$$

$$\left(\frac{25x^{4+6}}{y^{3+5}}\right)^{3/2}$$

$$\left(\frac{25x^{10}}{y^8}\right)^{3/2}$$

$$\frac{25^{\frac{3}{2}}x^{10\cdot\frac{3}{2}}}{y^{8\cdot\frac{3}{2}}}$$

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Simplify the expression

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$$\left(\frac{25x^{4+6}}{y^{3+5}}\right)^{3/2}$$

$$\left(\frac{25x^{10}}{y^8}\right)^{3/2}$$

$$\frac{25^{\frac{3}{2}}x^{10\cdot\frac{3}{2}}}{y^{8\cdot\frac{3}{2}}}$$

$$\frac{25^{\frac{3}{2}}x^{\frac{30}{2}}}{y^{\frac{24}{2}}}$$

$$\frac{25^{\frac{3}{2}}x^{15}}{y^{12}}$$

$$\frac{\left(\sqrt{25}\right)^3x^{15}}{y^{12}}$$

$$\frac{(5)^3x^{15}}{y^{12}}$$

$$\boxed{\frac{125x^{15}}{y^{12}}}$$

ans

Rationalize the denominator

$$\frac{3x}{\sqrt{x}} \cdot \frac{\sqrt{x}}{\sqrt{x}}$$

$$\frac{3x\sqrt{x}}{\left(\sqrt{x}\right)^2} \longrightarrow \frac{3x\sqrt{x}}{\left(x^{\frac{1}{2}}\right)^2}$$

$$\frac{3x\sqrt{x}}{x}$$

Rationalize the denominator

$$\frac{3x}{\sqrt{x}} \cdot \frac{\sqrt{x}}{\sqrt{x}}$$

$$\frac{3x\sqrt{x}}{\left(\sqrt{x}\right)^2} \longrightarrow \frac{3x\sqrt{x}}{\left(x^{\frac{1}{2}}\right)^2}$$

$$\frac{3x\sqrt{x}}{x}$$

$$\boxed{3\sqrt{x}}$$

ans

Simplify $\sqrt{60x^2y^6z^{-11}}$

Simplify $\sqrt{60x^2y^6z^{-11}}$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{\sqrt{60}\cdot\sqrt{x^2}\cdot\sqrt{y^6}}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{\sqrt{60}\cdot\sqrt{x^2}\cdot\sqrt{y^6}}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{\sqrt{60}\cdot x\cdot\sqrt{y^6}}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{\sqrt{60}\cdot x\cdot\sqrt{y^6}}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{\sqrt{60}\cdot x\cdot\sqrt{y^2\cdot y^2\cdot y^2}}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{\sqrt{60}\cdot x\cdot\sqrt{y^2}\cdot\sqrt{y^2}\cdot\sqrt{y^2}}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{\sqrt{60}\cdot x\cdot y\cdot y\cdot y}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{\sqrt{60}\cdot x\cdot y^3}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{\sqrt{60}\cdot x\cdot y^3}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{\sqrt{4 \cdot 15} \cdot x \cdot y^3}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{\sqrt{4}\cdot\sqrt{15}\cdot x\cdot y^3}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3}{\sqrt{z^2\cdot z^2\cdot z^2\cdot z^2\cdot z^2\cdot z}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3}{\sqrt{z^{11}}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3}{\sqrt{z^{10}\cdot z}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3}{\sqrt{z^{10}}\cdot\sqrt{z}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3}{(z^{10})^{\frac{1}{2}}\cdot\sqrt{z}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3}{z^5\cdot\sqrt{z}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3}{z^5\cdot\sqrt{z}}\cdot\frac{\sqrt{z}}{\sqrt{z}}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3}{z^5\cdot\sqrt{z}}\cdot\frac{\sqrt{z}}{\sqrt{z}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3\cdot\sqrt{z}}{z^5\cdot z}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3\cdot\sqrt{z}}{z^{5+1}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3\cdot\sqrt{z}}{z^6}$$

Simplify $\sqrt{60x^2y^6z^{-11}}$

$$\sqrt{\frac{60x^2y^6}{z^{11}}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3}{z^5\cdot\sqrt{z}}\cdot\frac{\sqrt{z}}{\sqrt{z}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3\cdot\sqrt{z}}{z^5\cdot z}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3\cdot\sqrt{z}}{z^{5+1}}$$

$$\frac{2\sqrt{15}\cdot x\cdot y^3\cdot\sqrt{z}}{z^6}$$

$$\boxed{\frac{2\sqrt{15z}\ xy^3}{z^6}}$$

ans

SEATWORK